DIY Brushed DC Motor

3-pole and 5-pole motors built by hand from a tin can, a bicycle spoke and copper wire

Nabiha Nowrin · Department of EEE, University of Chittagong · EEE 312 course project, May 2026

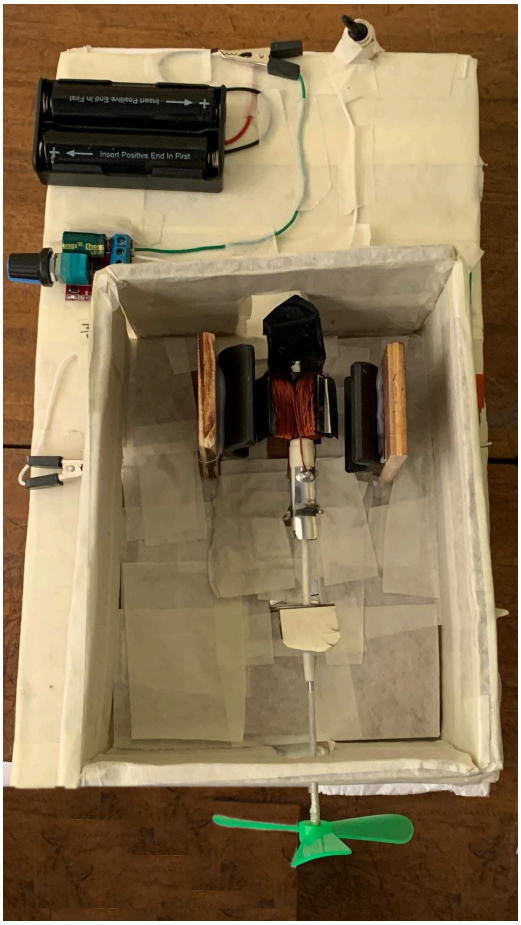


Figure 1. The finished motor with the lid open: battery holder and PWM controller on the base, fan on the shaft.

Project type	Team course project, EEE 312
Course teacher	Dr. Mohammad Arif Iftekhar Mahmood
Team	Nabiha Nowrin (main designer), Afrin Mostafa Riya, Minhaj Uddin Samrat
Motor	Brushed permanent-magnet DC motor, built in 3-pole and 5-pole versions
Core and winding	Tin-can core; 28 AWG enamelled copper wire
Commutator	Copper plate on an epoxy putty hub; iron wire brushes
Supply	Two 18650 Li-ion cells in series (7.4 V), through a PWM speed controller
Load	Fan blade
Award	Certificate of Achievement for excellence in DC motor design, Department of EEE, 20 May 2026

1 Overview

> **Same motor, one change: the number of poles.**
>
> The 3-pole motor needed a push to start. The 5-pole motor started on its own and ran more smoothly.

We built a working DC motor from everyday materials to understand how one turns electricity into rotation. The core was cut from a tin can, the shaft is a bicycle spoke, and the commutator is copper plate on epoxy putty. We made two armatures, one with 3 poles and one with 5, and kept everything else the same so we could compare them. A PWM module controls the speed, and a small fan acts as the load.

The motor won a Certificate of Achievement from the Department of EEE for excellence in DC motor design.

2 How it works

2.1 Force and torque

A wire carrying current inside a magnetic field is pushed sideways. The magnets give the field; the coils on the armature carry the current. The two sides of each coil are pushed in opposite directions, which turns the armature.

> $$F = B\,I\,L \qquad T = k\,\varphi\,I_{a}$$
>
> B = flux density (T), I = current (A), L = wire length in the field (m), φ = flux per pole, I_{a} = armature current, k = constant set by the turns and poles.

2.2 Commutator

Without help, a coil would stop once its sides passed the magnets. The commutator, a split copper ring on the shaft, fixes this: as it turns under the two fixed brushes, it reverses the current in each coil at the right moment, so the push always acts in the same direction.

2.3 Speed and PWM control

A spinning coil also acts as a generator and makes a back EMF that opposes the supply. Speed therefore rises with the voltage applied:

> $$\omega = (V - I_{a}R_{a}) \,/\, (k\,\varphi) \qquad V_{avg} = D \times V_{s}$$
>
> The PWM module switches the 7.4 V supply on and off very fast. At a 50% duty cycle (D) the motor sees 0.5 × 7.4 = 3.7 V on average, so turning the knob changes the speed.

2.4 Why more poles start better

Torque is not even through one turn. It drops when a coil sits between magnets or a brush crosses a gap in the commutator. With 3 poles these weak spots, or dead zones, are wide: if the rotor stops in one, it cannot start again by itself. With 5 poles, several coils push at every position, so the torque is smoother and the motor self-starts.

2.5 Why it heats up

The changing field induces small circulating currents inside the solid tin-can core (eddy currents), and the thin wire has resistance. Both turn energy into heat. Real motors use a laminated core, made of thin insulated steel sheets, to block the eddy currents.

2.6 The two designs

Feature	3-pole	5-pole
Winding	Concentrated (one coil on each pole)	Distributed (coils spread over more than one pole)
Turns	180	About 70

3 Build

3.1 Components

Part	Made from	Job
Winding	28 AWG enamelled copper wire	Becomes an electromagnet when current flows
Armature core	Tin can cut into a pole frame, taped	Holds the coils, guides the flux
Commutator	Copper plate on an epoxy putty (M-Seal) hub	Reverses the coil current
Brushes	Solid iron wire	Feed current to the commutator
Shaft	Bicycle spoke	Carries the rotor, commutator and fan
Magnets	Permanent magnets, close to the rotor	Fixed magnetic field
Supports and base	Tin-can strips on a wooden board	Hold the shaft and magnets
Speed control	PWM DC motor speed controller	Sets the average voltage
Supply	Two 18650 Li-ion cells in series; DC adapter	7.4 V DC

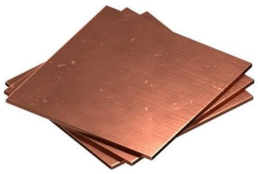
3.2 Materials



Enamelled copper wire



Tin can



Copper plate



Epoxy putty



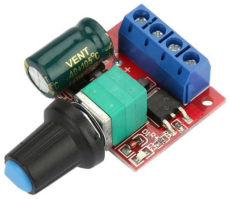
Solid iron wire



Bicycle spoke



Insulating tape



PWM speed controller



Rechargeable cells

Figure 2. Materials and parts used.

3.3 Steps

1. Cut the core from a tin can and wrap it in insulating tape.
2. Wind the enamelled wire on the poles (180 turns on the 3-pole, about 70 on the 5-pole).
3. Fix copper segments around an epoxy putty hub on the shaft and join the coil ends to them, scraping off the enamel first.
4. Mount the shaft on tin-can supports on the wooden base, with the magnets on both sides.
5. Bend iron wire brushes to press on the commutator; wire the cells through the PWM module; fit the fan.

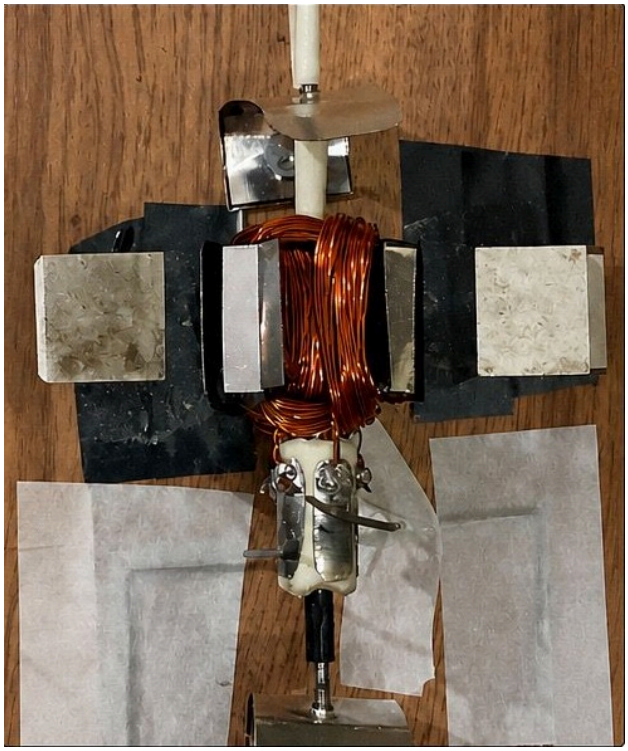


Figure 3. The motor outside its box: wound core in the middle, commutator below it, magnets on each side.

4 Results

	3-pole	5-pole
Starting	Needed a push (low starting torque)	Started on its own
Running	Smooth, moderate speed	Smooth, moderate speed
Dead zones	Present	Reduced

Both motors were built and ran on a low-voltage DC supply. As the theory predicts (Section 2.4), the 5-pole motor's overlapping coils removed the dead zones that stopped the 3-pole motor from starting. Fixing the commutator-to-core leakage improved efficiency.

5 Problems and fixes

Problem	Cause	Fix	Lesson
3-pole would not self-start	Dead zones with few poles	Built the 5-pole version	More poles, smoother torque
Heating	Eddy currents in the solid core; resistance of the thin wire	Not fixed	Use a laminated core
Vibration	Hand-made rotor not balanced	Not fixed	Balance the rotor on the shaft
Current leakage	Commutator touching the core	Fixed in the improved version	Insulate the commutator from the core
Neodymium magnets did not work	Not found	Used magnets from a ready-made DC motor	–

6 Award


UNIVERSITY OF CHITTAGONG
DEPARTMENT OF EEE
CERTIFICATE
OF ACHIEVEMENT
This is to certify that
Afrin Mostafa Riya, Nabiha Nowrin and Minhaz Uddin Samrat
have demonstrated exceptional creativity, technical knowledge and teamwork by achieving excellence in **DC motor design.**
We commend their outstanding achievement and wish them continued success in all their future endeavors.
20/05/2026
CHAIRMAN (in charge)
Department of EEE
University of Chittagong

Figure 4. Certificate of Achievement, Department of EEE, University of Chittagong, 20 May 2026.

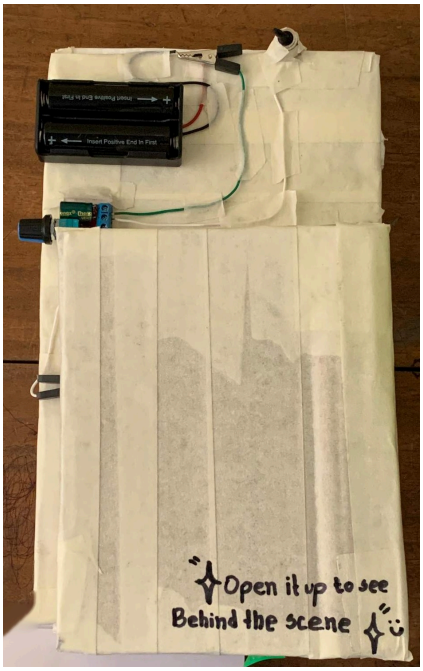


Figure 5. The motor on display, lid closed: "Open it up to see behind the scene".

7 Next steps

- Use a laminated core to cut heating.
- Improve the insulation between the commutator and the core.
- Balance the rotor to reduce vibration.
- Optimise the winding for more torque.

8 References

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